

What is a Neural Network?



Hi! I'm
Nelly the
learning robot!

🌟 An explanation for a 5-year-old 🌟



A neural network is like a team of tiny helpers that work together so a computer can learn and make smart guesses!



It learns by seeing lots of examples and learning from its mistakes.

1. Nelly Learns by Practice

1 First, Nelly sees a picture.



2 She makes a guess.



3 You tell her, "Not quite. That's a cat."



4 She tries again, changes a little bit, and learns from her mistake.



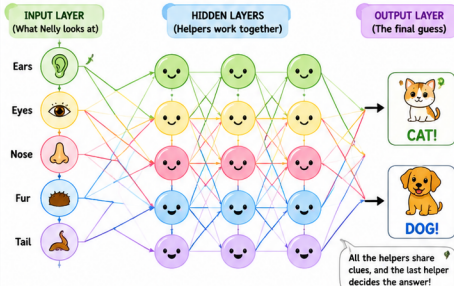
5 The more pictures she sees, the smarter she gets!



Every mistake helps Nelly turn her "knobs" a little bit. Those knobs are called **weights**.

2. How Nelly's Neural Network Works

Inside Nelly's brain, there are many tiny helpers called **neurons**.



Each helper looks at a small thing. Then they pass their clues to the next helpers. Little by little, they figure out the **big picture** together!

3. Training: Learn → Make Mistakes → Get Better!



Nelly sees lots of pictures.



She guesses right sometimes and wrong sometimes.



She learns from mistakes and turns her "knobs" (**weights**) a little bit.



The more she practices, the better and smarter she becomes!

4. Simple Summary

A neural network is like a classroom of tiny helpers that:



Notice small things



Share clues with each other



Learn from mistakes



Get better with practice



Make smart guesses!



In One Big Idea

Neural networks learn by looking, guessing, learning, and getting better—just like you learn every day!

